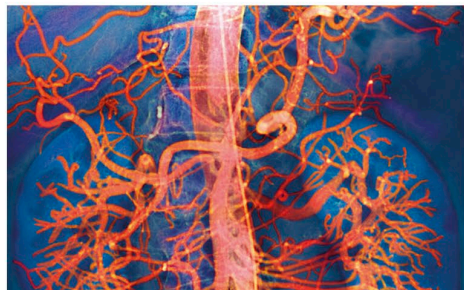


THE FLUID BODY

ROUGHLY TWO-THIRDS OF THE BODY IS COMPOSED OF WATER AND THE VARIOUS ESSENTIAL SUBSTANCES DISSOLVED IN IT. THESE FLUIDS ARE FOUND IN CELLS, AROUND THE BODY'S TISSUES, AND, MOST OBVIOUSLY, IN BLOOD AND LYMPH.



CIRCULATORY NETWORK

Blood is the fastest-circulating “fluid” in the body. Its liquid component, plasma, is constantly exchanging fluids with other body systems and structures.

There are about 40 litres (70 pints) of water in the average adult body, and it makes up the major part of most body parts. Tissues are 70–80 per cent water; blood plasma is more than 90 per cent; bones contain almost 25 per cent; and fat is 10–15 per cent water.

TYPES AND FUNCTIONS OF FLUIDS

There are two major categories of body fluids – intracellular and extracellular. Intracellular fluid (also called cytoplasm) is found inside cells. Extracellular fluid accounts for all other fluids in the body. Its subcategories are: interstitial fluid in the spaces between cells and tissues; blood plasma and lymph; the fluids in bones, joints, and dense connective tissue; and transcellular fluid, which includes saliva, mucus, sweat, and urine.

Water is an excellent solvent and the thousands of substances dissolved in it are used in the biochemical reactions that are the very basis of life. Water also distributes

CARDIOVASCULAR SYSTEM

144–159

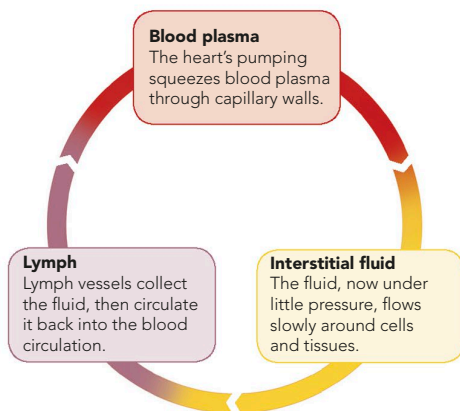
LYMPH AND IMMUNE SYSTEMS

190–209

nutrients around the body and collects and delivers wastes. Fluids spread heat from active areas to cooler ones, and act as shock absorbers to cushion sensitive areas such as the brain. Fluids also work as lubricants, so that tissues and organs slip past each other with minimal friction.

BLOOD AND LYMPH

The blood and lymphatic circulatory systems are constantly swapping fluids (see the illustration below). Blood plasma transports red blood cells, white blood cells, platelets, and a wide variety of nutrients and chemicals around the body. Lymph fluid carries white blood cells and other substances, such as fats and proteins.



BLOOD PLASMA AND LYMPH CYCLE

Blood plasma leaks from capillaries to form interstitial fluid. Some of this drains into lymph vessels to become lymph fluid, which then returns to the blood circulation.